2012

6 **4** **24**

1 4

A a^2+a^2 a^4　　B $a^6 \div a^2$ a^3　　C $a \bullet a^2$ a^3　　D a^2 ^{3} a^5

2 4 a b c 0

A $a+c$ $b+c$　　B $a+c$ $b+c$　　C ac bc　　D $\frac{a}{c}<\frac{b}{c}$

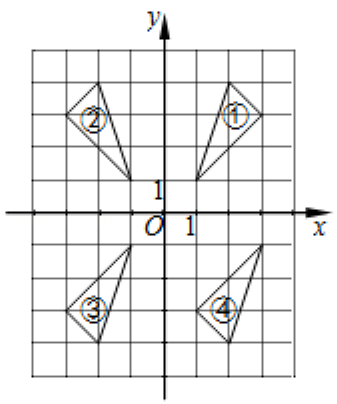
3 4 y x 2

A　　B　　C　　D

4 4 2 3 2 3 2 3 3 2 $-\frac{3}{2}$ 4

A 2 3　　B 2 3　　C 2 3　　D $-\frac{3}{2}$ 4

5 4 ① ② ③ ④ x



A ① ②　　B ② ③　　C ① ③　　D ② ④

6 4

A

B

C

D

12 **4** **48**

7　4　　　a　$2b$　$a+b$　______

8　4　　$\frac{1}{x}-\frac{1}{x+1}$　______

9　4　　x　　x^2+kx+9　0　k　　　　　　k　______

10　4　　$f(x)=\sqrt{x+6}$　f　a　a　a　______

11　4　　　　　y　　　　　y　　　　　A　1　1

12　4　　　　　2　　n　　　　　　　　　　　　　　$\frac{4}{5}$　n

______

13　4　　2　　60°　　　　______

14　4　$\triangle ABC$　D　E　　AB　AC　　$DE \parallel BC$　AD　5　DB　10　$S_{\triangle ADE}$　$S_{\triangle ABC}$

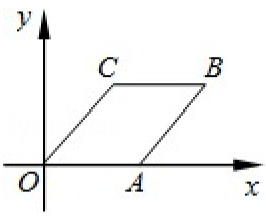
______

15　4　$\triangle ABC$　$\angle A$　90°　$\angle B$　θ　AC　b　AB　______　b　θ

16　4　G　$\triangle ABC$　　$\overrightarrow{AB}=\vec{a}$　$\overrightarrow{AC}=\vec{b}$　$\overrightarrow{AG}$　______　$\vec{a}$　$\vec{b}$

17　4　$\odot O_1$　$\odot O_2$　$\odot O_1$　$\odot O_2$　2　1　O_1O_2　7　$\odot O_2$　　______

18　4　　　　A　x　　B　　4　2　　$OABC$　　C　　______



□□□□□□□□□□□□7□□□□78□□

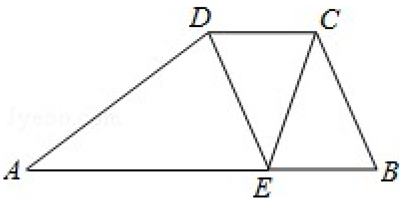
19　10　　$(-3)^0-27^{\frac{1}{3}}+|1-\sqrt{2}|+\sqrt{2}\times\sqrt{6}+(\sqrt{3}+\sqrt{2})^{-1}$

20　10　$\begin{cases} x^2-9y^2=0 \\ x^2-2xy+y^2=4 \end{cases}$

21　10　　　$ABCD$　$AB \parallel CD$　AB　13　CD　4　E　AB　$DE \parallel BC$

1　CE　CB　$\tan\angle B$　3　$\triangle ADE$

2　$\angle DEC$　$\angle A$　BC



22　10　$\odot O_1$ $\odot O_2$　T　T　$\odot O_1$ $\odot O_2$　A B

1　$\odot O_1$ $\odot O_2$　1　AT BT

2　$\odot O_1$ $\odot O_2$　R r　2　AT BT　R r

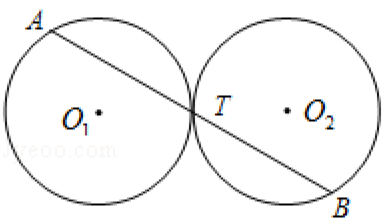


图 1

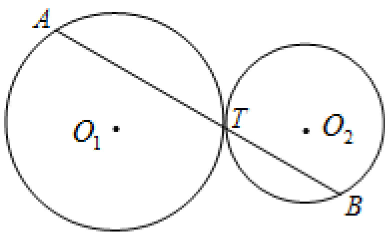


图 2

23　12　“　”　600　“　”

98

1

	x 60	$60\le x$ 80	$x\ge 80$
		40	
	57	a	b

1　______

2

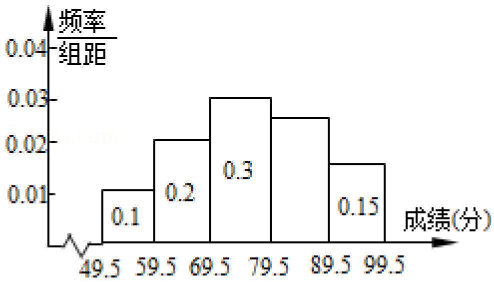
3　76.5　1　b　a　15　a b

4　$p\le x\le q$　x　$[p$ $q]$　1　a　______

A [69.5 79.5] B [65 74]

C [66.5 75.5] D [66 75]

抽样分析频率分布直方图



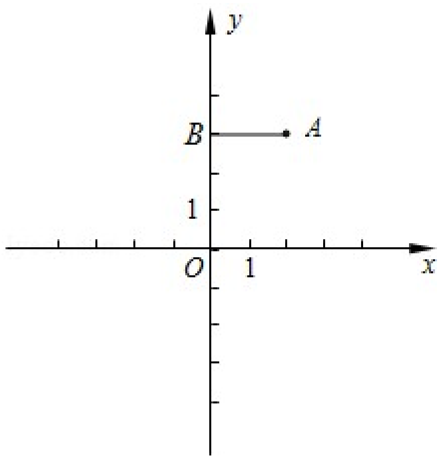
24 12 xOy A 2 3 AB y B AB A

90° B C BC x D

1 D

2 A B D E

3 2 F A E F $\triangle ACD$

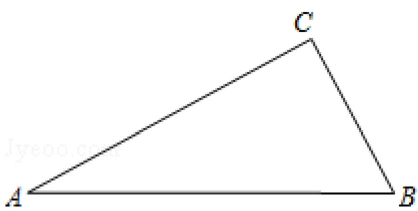


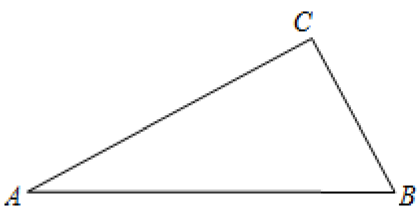
25 14 $\triangle ABC$ $\angle ACB$ 90° P $\angle ACB$ PA PB

1 P $\triangle ABP$

2 PA m PC n m n $\triangle ABC$

3 CP AB D AC BC $\frac{CD}{AC}+\frac{CD}{BC}$





备用图

2012

6 4 24

1 4

A a^2+a^2 a^4 B $a^6\div a^2$ a^3 C $a\bullet a^2$ a^3 D a^2 3 a^5

35 46 47 48

11

A a^2+a^2 $2a^2$

B $a^6\div a^2$ a^4

C $a\bullet a^2$ a^3

D a^2 3 a^6

C

2 4 a b c 0

A $a+c$ $b+c$ B $a+c$ $b+c$ C ac bc D $\frac{a}{c}<\frac{b}{c}$

C2

A *c*

B 1 *c*

C *c*

D *c*

A

1

2

3

3 4 y x 2

A B C D

F5

k 0 b 0 y

y x 2

∵k 1 0

∴

∵b 2 0

∴y

∴

B

y $kx+b$ k 0 k 0

4 4 2 3 2 3 2 3 3 2 $-\frac{3}{2}$ 4

A 2 3 B 2 3 C 2 3 D $-\frac{3}{2}$ 4

G6

k k xy

A 2× 3 6

B 2 ×3 6

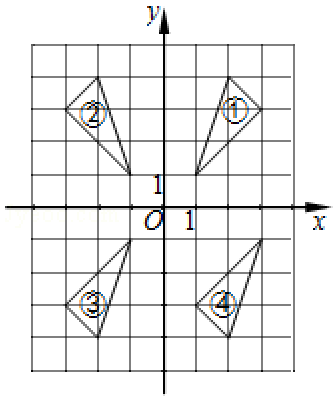
C 2 × 3 6

D $\frac{3}{2}$ ×4 6

C

y $\frac{k}{x}$ k

5 4 ① ② ③ ④ x



A ① ② B ② ③ C ① ③ D ② ④

P2

x

x ② ③

x ② ③

B

x

6 4

A

B

C

D

O1

A

B

C

D

A d r

B

C

D

D

□□□□□□□□□□□□**12**□□□□**4**□□□□**48**□□

7　4　　a $2b$　$a+b$　<u>a^2 ab $2b^2$</u>

4B

a^2+ab $2ab$ $2b^2$ a^2 ab $2b^2$

a^2 ab $2b^2$

8　4　　$\frac{1}{x}-\frac{1}{x+1}$　<u>$\frac{1}{x(x+1)}$</u>

6B

$\frac{x+1-x}{x(x+1)}$ $\frac{1}{x(x+1)}$

9　4　　x　x^2+kx+9 0 k　　k <u>±6</u>

AA

2B

x　x^2+kx+9 0 k　　△ 0　k

x　x^2+kx+9 0 k

∴△ k^2 4×9 k^2 36 0　k ±6

±6

k

10　4　　$f(x)=\sqrt{x+6}$　f a　a　a <u>3</u>

E5

a　x　a

f a　a

∴$\sqrt{a+6}$ a

a 3　2

3

1　2

11 4 y y A 1 1

y x^2+2

H3

26

y y

y a 0 b 0 A 1 1

y ax^2+bx+c $a\neq0$

∵ y y

∴ y a 0 b 0

∵ A 1 1

∴ $a+c$ 1

∴ a 1 c 2

∴ y x^2+2

y x^2+2

a

12 4 2 n $\frac{4}{5}$ n

8

X4

$n+2$ n

P $\frac{n}{n+2}$ $\frac{4}{5}$

n 8

8

n A m A P A $\frac{m}{n}$

13 4 2 60° 2

MN

17

$\triangle OAB$ AB $\odot O$ OA

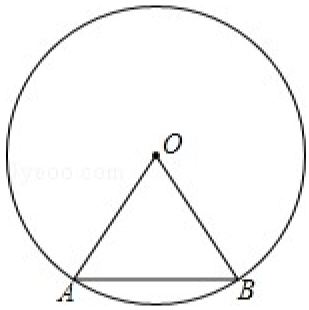
OA 2 $\angle OAB$ 60°

$\because OA$ OB

$\therefore \triangle OAB$

$\therefore AB$ OA 2

2



“ 60 ”

14 4 $\triangle ABC$ D E AB AC $DE \parallel BC$ AD 5 DB 10 $S_{\triangle ADE}$ $S_{\triangle ABC}$

$\frac{1}{9}$

S9

11

$\triangle ADE \backsim \triangle ABC$

$DE \parallel BC$

$\therefore \angle ADE$ $\angle B$ $\angle AED$ $\angle C$

$\therefore \triangle ADE \backsim \triangle ABC$

$\because AD$ 5 DB 10

$\therefore \frac{AD}{AE}$ $\frac{5}{5+10}$ $\frac{1}{3}$

$S_{\triangle ADE}$ $S_{\triangle ABC}$ $\frac{1}{3}$ 2 $\frac{1}{9}$

$\frac{1}{9}$

15　4　　$\triangle ABC$　$\angle A$ 90°　$\angle B$ θ AC b　AB ___b•cotθ___　b　θ

T7

11

$\triangle ABC$　$\angle A$ 90° BC

∴AB AC•cot$\angle B$ b•cotθ

16　4　　G $\triangle ABC$　　$\overrightarrow{AB}=\vec{a}$ $\overrightarrow{AC}=\vec{b}$　$\overrightarrow{AG}$ ___$\frac{1}{3}(\vec{a}+\vec{b})$___　$\vec{a}$ $\vec{b}$

K5　　LM *

$\overrightarrow{BC}$　G $\triangle ABC$　$\overrightarrow{AD}$　$\overrightarrow{AG}$

∵$\overrightarrow{AB}=\vec{a}$ $\overrightarrow{AC}=\vec{b}$

∴$\overrightarrow{BC}$ $\overrightarrow{AC}$ $\overrightarrow{AB}$ $\vec{b}$ $\vec{a}$

∵D BC

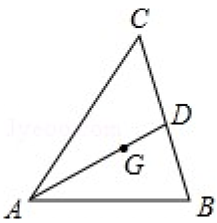
∴$\overrightarrow{BD}$ $\frac{1}{2}\overrightarrow{BC}$ $\frac{1}{2}$ $\vec{b}$ $\vec{a}$

∴$\overrightarrow{AD}$ $\overrightarrow{AB}+\overrightarrow{BD}$ $\vec{a}+\frac{1}{2}$ $\vec{b}$ $\vec{a}$ $\frac{1}{2}$ $\vec{a}+\vec{b}$

∵G $\triangle ABC$

∴$\overrightarrow{AG}$ $\frac{2}{3}\overrightarrow{AD}$ $\frac{2}{3}\times\frac{1}{2}$ $\vec{a}+\vec{b}$ $\frac{1}{3}$ $\vec{a}+\vec{b}$

$\frac{1}{3}$ $\vec{a}+\vec{b}$



17　4　　⊙O_1 ⊙O_2　⊙O_1　⊙O_2　2　1　O_1O_2 7　⊙O_2　___2　6___

MK

11

$\odot O_2$ r $\odot O_1$ R $2r+1$ $\odot O_1$ $\odot O_2$ r

r

$\odot O_2$ r $\odot O_1$ R $2r+1$

$\odot O_1$ $\odot O_2$ $r+R$ O_1O_2 7 **R** r O_1O_2 7

$3r+1$ 7 $r+1$ 7

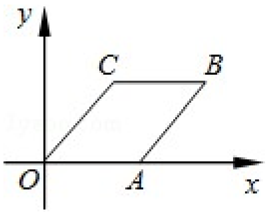
r 2 6

2 6

d **R** r d **R**$+r$

18 4 A x B 4 2 $OABC$ C ___

$\frac{3}{2}$， 2___

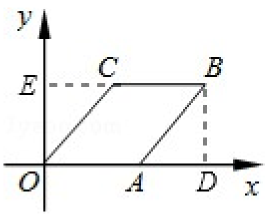


D5 L8

$OABC$ OC OA AB BC $BC\parallel OA$ B $BD\perp OA$ D BC

y E AB x OA x AD 4 x Rt△ABD BC OD C

B $BD\perp OA$ D BC y E



∵ $OABC$

∴OC OA AB BC $BC\parallel OA$

AB x OA x AD 4 x

Rt△ABD AB^2 AD^2+BD^2

x^2 4 x $^2+2^2$

$x \quad \frac{5}{2}$

$\therefore CE \quad BE \quad BC \quad OD \quad BC \quad 4 \quad \frac{5}{2} \quad \frac{3}{2}$

$\therefore C \qquad \frac{3}{2} \quad 2$

$\frac{3}{2} \quad 2$

□□□□□□□□□□□□□**7**□□□□**78**□□

19　10　$(-3)^{0}-27^{\frac{1}{3}}+|1-\sqrt{2}|+\sqrt{2}\times\sqrt{6}+(\sqrt{3}+\sqrt{2})^{-1}$

2C　2F　6E　6F

4

$1-3+\sqrt{2}-1+2\sqrt{3}+\frac{1}{\sqrt{3}+\sqrt{2}}$

$1-3+\sqrt{2}-1+2\sqrt{3}+\sqrt{3}-\sqrt{2}$

$3\sqrt{3}-3$

20　10　$\begin{cases}x^{2}-9y^{2}=0\\x^{2}-2xy+y^{2}=4\end{cases}$

AF

$\begin{cases}(x-3y)(x+3y)=0\\(x-y-2)(x-y+2)=0\end{cases}$　$\begin{cases}x-3y=0\\x-y-2=0\end{cases}$ $\begin{cases}x-3y=0\\x-y+2=0\end{cases}$ $\begin{cases}x+3y=0\\x-y-2=0\end{cases}$

$\begin{cases}x+3y=0\\x-y+2=0\end{cases}$　4

$\begin{cases}(x-3y)(x+3y)=0\\(x-y-2)(x-y+2)=0\end{cases}$

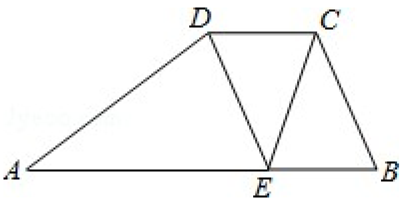
$\begin{cases}x-3y=0\\x-y-2=0\end{cases}$ $\begin{cases}x-3y=0\\x-y+2=0\end{cases}$ $\begin{cases}x+3y=0\\x-y-2=0\end{cases}$ $\begin{cases}x+3y=0\\x-y+2=0\end{cases}$

$$\begin{cases}x_1=3\\y_1=1\end{cases}\quad\begin{cases}x_2=-3\\y_2=-1\end{cases}\quad\begin{cases}x_3=\frac{3}{2}\\y_3=-\frac{1}{2}\end{cases}\quad\begin{cases}x_4=-\frac{3}{2}\\y_4=\frac{1}{2}\end{cases}$$

21 10 *ABCD* *AB*∥*CD* *AB* 13 *CD* 4 *E* *AB* *DE*∥*BC*

1 *CE* *CB* tan∠*B* 3 △*ADE*

2 ∠*DEC* ∠*A* *BC*



LH S9 T7

1 *C* *D* *CF*⊥*AB* *DG*⊥*AB* *AB* *F* *G* *CDEB* *AE*

Rt△*BCF* △*ADE*

2 △*CDE*∽△*DEA* *DE* *BC*

1 *C* *D* *CF*⊥*AB* *DG*⊥*AB* *AB* *F* *G*

∵*AB*∥*CD*

∴*DG* *CF*

∵*AB*∥*CD* *DE*∥*BC*

∴ *CDEB*

∴*BE* *CD*

∵*AB* 13 *CD* 4

∴*AE* *AB* *BE* 13 4 9

∵*CE* *CB* *CF*⊥*BE*

∴*BF* $\frac{1}{2}$*BE* $\frac{1}{2}$×4 2

Rt△*BCF* tan∠*B* 3 *BF* 2

∴tan∠*B* $\frac{CF}{BF}$ 3

$\frac{CF}{2}$ 3 *CF* 6

∴*DG* *CF* 6

$\therefore S_{\triangle ADE}\ \frac{1}{2}AE\bullet DG\ \frac{1}{2}\times 9\times 6\ \ 27$

2 $\because AB\parallel CD$

$\therefore\angle CDE\ \ \angle DEA$

$\because\angle DEC\ \ \angle A$

$\therefore\triangle CDE\backsim\triangle DEA$

$\therefore\frac{CD}{DE}=\frac{DE}{EA}$

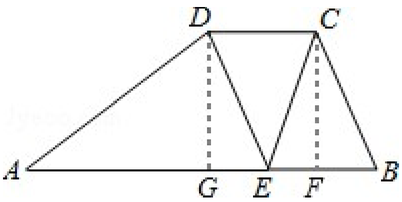
$\because AE\ \ 9\ \ CD\ \ 4$

$\therefore\frac{4}{DE}=\frac{DE}{9}$

$\therefore DE^2\ \ 36\ \ DE\ \ 6$

$\because\ CDEB$

$\therefore BC\ \ DE\ \ 6$



22 10 $\odot O_1$ $\odot O_2$ T T $\odot O_1$ $\odot O_2$ A B

1 $\odot O_1$ $\odot O_2$ 1 AT BT

2 $\odot O_1$ $\odot O_2$ R r 2 AT BT R r

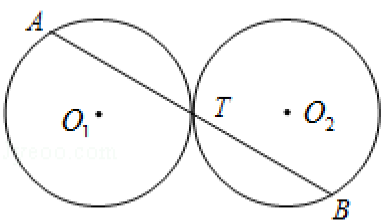


图 1

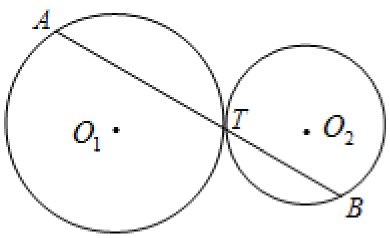


图 2

MR

14

1 O_1O_2 1 O_1O_2 T

$\triangle O_1CT$ $\triangle O_2DT$ CT DT

$O_1C \perp AT$ CT AT DT BT AT BT

2 AT BT R r $\frac{AT}{BT}$ $\frac{R}{r}$ O_1O_2 2

O_1O_2 T $\triangle O_1CT$ $\triangle O_2DT$

O_1T R O_2T r CT DT R r $O_1C \perp AT$ CT AT

DT BT AT BT R r

1 O_1O_2 1

$\because \odot O_1$ $\odot O_2$ T

$\therefore T$ O_1O_2

O_1 O_2 $O_1C \perp AT$ $O_2D \perp BT$ C D

$\therefore \angle O_1CT$ $\angle O_2DT$ $90°$ $\angle O_1TC$ $\angle O_2TD$

$\therefore \triangle O_1CT \backsim \triangle O_2DT$

$\therefore \frac{CT}{DT}$ $\frac{O_1T}{O_2T}$

$\because \odot O_1$ $\odot O_2$

$\therefore O_1T$ O_2T

$\therefore \frac{CT}{DT}$ $\frac{O_1T}{O_2T}$ 1

$\therefore CT$ DT

$\odot O_1 \because O_1C \perp AB$

$\therefore AC$ CT $\frac{1}{2}AT$

BD DT $\frac{1}{2}BT$

$\therefore \frac{1}{2}AT$ $\frac{1}{2}BT$ AT BT

2 AT BT R r $\frac{AT}{BT}$ $\frac{R}{r}$

1 O_1O_2 2

$\because \odot O_1$ $\odot O_2$ T

$\therefore T \quad O_1O_2$

$O_1 \ O_2 \quad O_1C \perp AT \ O_2D \perp BT \quad C \ D$

$\therefore \angle O_1CT \ \angle O_2DT \ 90° \quad \angle O_1TC \ \angle O_2TD$

$\therefore \triangle O_1CT \backsim \triangle O_2DT$

$$\therefore \frac{CT}{DT} \ \frac{O_1T}{O_2T}$$

$\because O_1T \ R \ O_2T \ r$

$$\therefore \frac{CT}{DT} \ \frac{O_1T}{O_2T} \ \frac{R}{r}$$

$\odot O_1 \because O_1C \perp AB$

$\therefore AC \ CT \ \frac{1}{2}AT$

$BD \ DT \ \frac{1}{2}BT$

$$\therefore \frac{AT}{BT} \ \frac{\frac{1}{2}AT}{\frac{1}{2}BT} \ \frac{CT}{DT} \ \frac{R}{r}$$

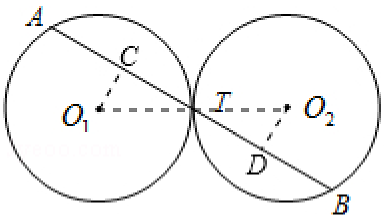


图 1

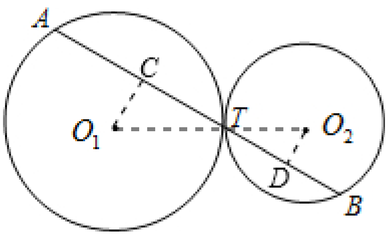


图 2

23 12 “ ” 600 “ ”

98

1

	$x \ 60$	$60 \leq x \ 80$	$x \geq 80$

		40	
	57	*a*	*b*

1 80

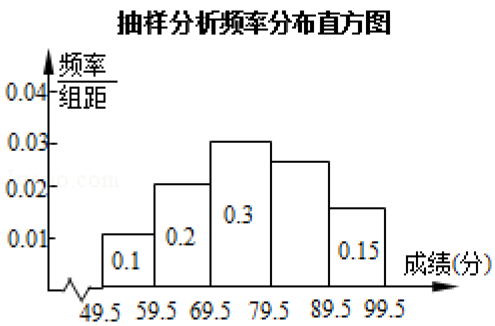
2

3 76.5 1 *b* *a* 15 *a* *b*

4 $p \le x \le q$ x $[p\ q]$ 1 *a* D

A [69.5 79.5] *B* [65 74]

C [66.5 75.5] *D* [66 75]



V5 V7 V8 W2

27

1 40

2 1 600

3 *b* *a* 15 *a* *b*

4 *a*

1 0.2+0.3 0.5

40÷0.5 80 … 3

2 79.5 89.5 1 0.1+0.2+0.3+0.15 0.25 … 1

0.25+0.15 0.4

600×0.4 240 … 2

3 80×0.1 8

80×0.4 32 … 1

$$\begin{cases}\frac{57\times 8+40a+32b}{80}=76.5\\ -a+b=15\end{cases}$$ … 1

$$\begin{cases}a=72\\ b=87\end{cases}$$ … 1

a *b* 72 87

4 59.5 69.5 80×0.2 16

69.5 79.5 80×0.3 24

∵

$\therefore a\geq\frac{16\times 60+24\times 70}{16+24}$ $\frac{2640}{40}$ 66

$a\leq\frac{16\times 69+24\times 79}{16+24}$ $\frac{3000}{40}$ 75

a $66\leq a\leq 75$

[66 75]

D … 3

4

a

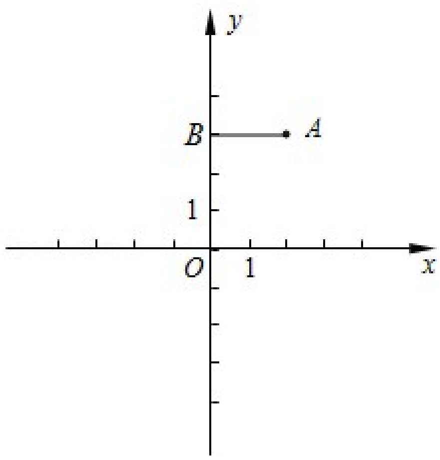
24 12 *xOy* *A* 2 3 *AB* *y* *B* *AB* *A*

90° *B* *C* *BC* *x* *D*

1 *D*

2 *A* *B* *D* *E*

3 2 *F* *A* *E* *F* △*ACD*



HF

15 32

1 A AB AB C BC

D

2 E

3 A B C D $\angle ACD$ 135° A E F

F

1 C 2 1

BC y $mx+n$ $m\neq0$

$$\begin{cases} n=3 \\ 2m+n=1 \end{cases}$$

$$\begin{cases} m=-1 \\ n=3 \end{cases}$$

BC y $x+3$

y 0 0 $x+3$ x 3

D 3 0

2 A B D

y ax^2+bx+c $a\neq0$

$$\begin{cases} 4a+2b+c=3 \\ c=3 \\ 9a+3b+c=0 \end{cases}$$

$$\begin{cases} a=-1 \\ b=2 \\ c=3 \end{cases}$$

y x^2+2x+3

E 1 4

3 F y x^2+2x+3 x 1 F 1 m

AB AC $\angle BAC$ 90°

∴$\angle ACB$ 45° $\angle ACD$ 180° $\angle ACB$ 135°

A E F $\triangle ACD$ $\triangle AEF$ 135°

F E $\angle AEF$ $\angle ACD$ 135° EF m 4

$\frac{EF}{CA}$ $\frac{EA}{CD}$ $\frac{EF}{CA}$ $\frac{CD}{EA}$

A E F $\triangle ACD$

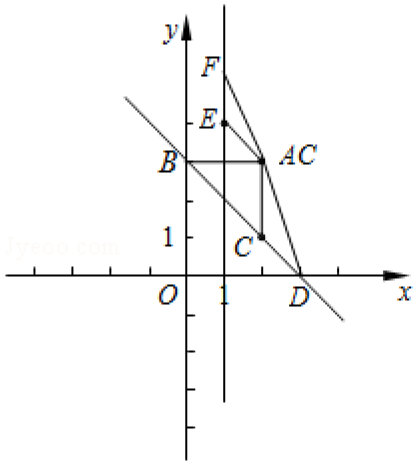
D 3 0 C 2 1 A 2 3 E 1 4

AC 3 1 2 CD $\sqrt{2}$ AE $\sqrt{2}$

$\therefore \frac{m-4}{2}$ $\frac{\sqrt{2}}{\sqrt{2}}$ $\frac{m-4}{\sqrt{2}}$ $\frac{\sqrt{2}}{2}$

m 6 m 5

F 1 5 1 6



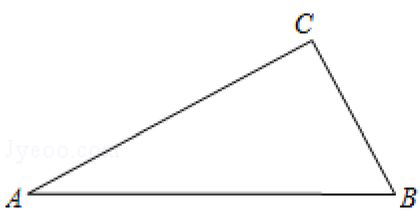
3

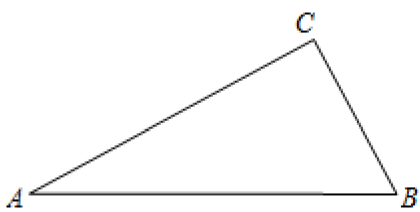
25 14 $\triangle ABC$ $\angle ACB$ 90° P $\angle ACB$ PA PB

1 P $\triangle ABP$

2 PA m PC n m n $\triangle ABC$

3 CP AB D AC BC $\frac{CD}{AC}+\frac{CD}{BC}$





备用图

SO

2B

1 P P $PE\perp AC$ $PF\perp CB$ E F

Rt△APE≌Rt△BPF △ABP

2 Rt△PAB $\angle APB$ 90° PA PB PA m AB $\sqrt{2}m$ Rt△APE≌Rt△BPF

△PCE≌△PCF AE BF CE CF $CA+CB$ $CE+EA+CB$ $CE+CF$ $2CE$

Rt△PCE $\angle PEC$ 90° $\angle PCE$ 45° PC n CE PE $\frac{\sqrt{2}}{2}n$ $CA+CB$ $2CE$ $\sqrt{2}n$

△ABC $AB+BC+CA$ $S_{\triangle ABC}$ $S_{\triangle PAC}+S_{\triangle PBC}$ $S_{\triangle PAB}$

3 D $DM\perp AC$ $DN\perp BC$ M N DM DN

$CD\sin 45°$ $\frac{\sqrt{2}}{2}CD$ $\frac{DN}{AC}$ $\frac{DB}{AB}$ $\frac{DM}{BC}$ $\frac{AD}{AB}$

1 P $\angle ACB$ AB

1 $\angle ACB$ CP AB PM CP PM P

△ABP

P $PE\perp AC$ $PF\perp CB$ E F 2

∵PC $\angle ACB$ $PE\perp AC$ $PF\perp CB$ E F

∴PE PF

Rt△APE Rt△BPF

∵$\begin{cases} PE=PF \\ PA=PB \end{cases}$

∴Rt△APE≌Rt△BPF

∴$\angle APE$ $\angle BPF$

∵$\angle PEC$ 90° $\angle PFC$ 90° $\angle ECF$ 90°

∴$\angle EPF$ 90°

∴$\angle APB$ 90°

∵PA PB

∴△ABP

2 2∵ Rt△PAB $\angle APB$ 90° PA PB PA m

∴AB $\sqrt{2}m$

Rt△APE≌Rt△BPF △PCE≌△PCF AE BF CE CF

$\therefore CA+CB\ \ CE+EA+CB\ \ CE+CF\ \ 2CE$

$Rt\triangle PCE\quad \angle PEC\ \ 90°\ \ \angle PCE\ \ 45°\ \ PC\ \ n$

$\therefore CE\ \ PE\ \ \frac{\sqrt{2}}{2}n$

$\therefore CA+CB\ \ 2CE\ \ \sqrt{2}n$

$\therefore \triangle ABC\qquad AB+BC+CA\ \ \sqrt{2}m+\sqrt{2}n$

$\because S_{\triangle ABC}\ \ S_{\triangle PAC}+S_{\triangle PBC}\ \ S_{\triangle PAB}$

$\frac{1}{2}AC\bullet PF+\frac{1}{2}BC\bullet PF\ \ \frac{1}{2}PA\bullet PB$

$\frac{1}{2}\ \ AC+BC\ \ \bullet PE\ \ \frac{1}{2}PA^2$

$\frac{1}{2}\times\sqrt{2}n\times\frac{\sqrt{2}}{2}n\ \ \frac{1}{2}m^2$

$\frac{1}{2}n^2\ \ \frac{1}{2}m^2\ \ n\ \ m$

$[\ \ S_{\triangle ABC}\ \ \frac{1}{2}AC\bullet BC\ \ \frac{1}{4}[\ \ AC+BC\ \ ^2\quad AC^2+BC^2\ \ ]\ \ \frac{1}{2}\ \ n^2\ \ m^2\ \ ]$

3

1 $\quad D\quad DM\perp AC\ \ DN\perp BC\quad M\ \ N$ 3

$DM\ \ DN\ \ CD\sin 45°\ \ \frac{\sqrt{2}}{2}CD$

$DN\parallel AC\ \ \frac{DN}{AC}\ \ \frac{DB}{AB}$①

$DM\parallel BC\ \ \frac{DM}{BC}\ \ \frac{AD}{AB}$②

①+② $\ \ \frac{DN}{AC}+\frac{DM}{BC}\ \ \frac{DB+AD}{AB}\quad \frac{DN}{AC}+\frac{DM}{BC}\ \ 1$

$\therefore\frac{\sqrt{2}}{2}\ \ \frac{CD}{AC}+\frac{CD}{BC}\quad 1\quad \frac{CD}{AC}+\frac{CD}{BC}\ \ \sqrt{2}$

2 $\quad$ 1$\because S_{\triangle ABC}\ \ S_{\triangle ACD}+S_{\triangle BCD}\ \ S_{\triangle ABC}\ \ \frac{1}{2}AC\bullet BC$

$\therefore S_{\triangle ACD}+S_{\triangle BCD}\ \ \frac{1}{2}AC\bullet DM+\frac{1}{2}BC\bullet DN\ \ \frac{1}{2}\ \ AC+BC\ \ \bullet\frac{\sqrt{2}}{2}CD$

$\therefore \frac{1}{2}$ $AC+BC$ $\bullet \frac{\sqrt{2}}{2}CD$ $\frac{1}{2}AC \bullet BC$

$\therefore \frac{(AC+BC)CD}{AC \bullet BC}$ $\sqrt{2}$ $\frac{CD}{AC}+\frac{CD}{BC}$ $\sqrt{2}$

3 D $DN \perp BC$ N 4

Rt CDN $\angle DCN$ $45°$ DN CN $\frac{\sqrt{2}}{2}CD$

$DN \parallel AC$ $\frac{DN}{AC}$ $\frac{DB}{AB}$① $\frac{CN}{BC}$ $\frac{AD}{AB}$②

①+② $\frac{DN}{AC}+\frac{CN}{BC}$ $\frac{DB+AD}{AB}$ $\frac{DN}{AC}+\frac{CN}{BC}$ 1

$\frac{\sqrt{2}}{2}$ $\frac{CD}{AC}+\frac{CD}{BC}$ 1 $\frac{CD}{AC}+\frac{CD}{BC}$ $\sqrt{2}$

4 B $BG \parallel DC$ AC G 5

$\angle G$ $\angle ACD$ $\angle BCD$ $\angle CBG$ $45°$ BG $\sqrt{2}BC$ $\sqrt{2}CG$

$\because BG \parallel DC$

$\therefore \frac{BG}{CD}$ $\frac{AG}{AC}$

$\therefore \frac{\sqrt{2}BC}{CD}$ $\frac{AC+BC}{AC}$ $\frac{(AC+BC)\bullet CD}{AC \bullet BC}$ $\sqrt{2}$

$\frac{CD}{AC}+\frac{CD}{BC}$ $\sqrt{2}$

5 A CB CD K 6

CK $\sqrt{2}AC$ DK CK CD $\sqrt{2}AC$ CD

$\frac{CD}{BC}$ $\frac{DK}{AK}$ $\frac{CD}{BC}$ $\frac{\sqrt{2}AC-CD}{AC}$

$\frac{CD}{BC}$ $\sqrt{2}$ $\frac{CD}{AC}$ $\frac{CD}{AC}+\frac{CD}{BC}$ $\sqrt{2}$

6 A B CD BC H AC G 7

AH $\sqrt{2}AC$ BG $\sqrt{2}BC$

$\therefore \frac{CD}{BG}$ $\frac{AD}{AB}$ $\frac{CD}{AH}$ $\frac{BD}{AB}$

$\therefore \frac{CD}{AH}+\frac{CD}{BG}$ 1

$\frac{CD}{\sqrt{2}AC}+\frac{CD}{\sqrt{2}BC}$ 1 $\frac{CD}{AC}+\frac{CD}{BC}$ $\sqrt{2}$

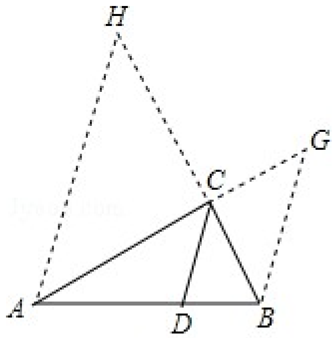


图7

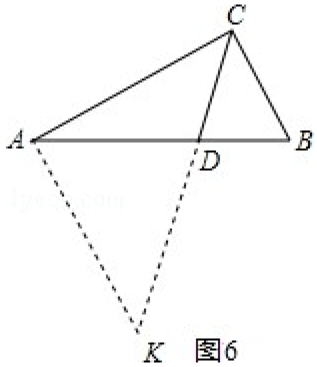


图6

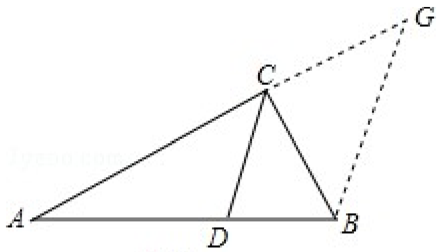


图5

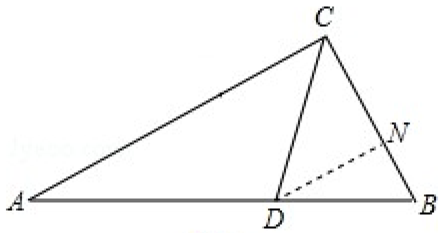


图4

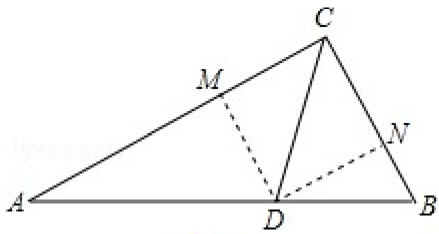


图3

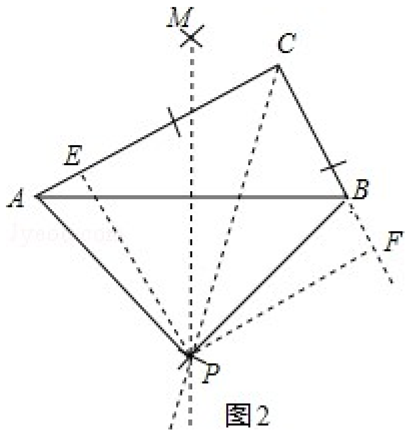


图2

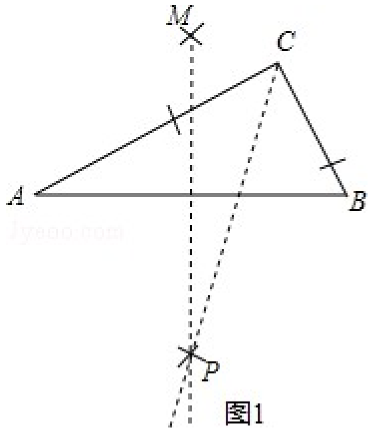


图1